

CSS methods & selectors

HTML Styles - CSS

- CSS stands for Cascading Style Sheets.
- CSS saves a lot of work. It can control the layout of multiple web pages all at once. It is used to format the layout of a webpage.
- With CSS, you can control the color, font, the size of text, the spacing between elements, how elements are positioned and laid out, what background images or background colors are to be used, different displays for different devices and screen sizes, and much more!

CSS

- The word **cascading** means that a style applied to a parent element will also apply to all children elements within the parent.
- So, if you set the color of the body text to "blue", all headings, paragraphs, and other text elements within the body will also get the same color (unless you specify something else)!

Using CSS

- CSS can be added to HTML documents in 3 ways:
 - **Inline** - by using the style attribute inside HTML elements
 - **Internal** - by using a `<style>` element in the `<head>` section
 - **External** - by using a `<link>` element to link to an external CSS file
- The most common way to add CSS, is to keep the styles in external CSS files.

Inline CSS

- An inline CSS is used to apply a unique style to a single HTML element.
- An inline CSS uses the style attribute of an HTML element.

```
<!DOCTYPE html>
<html>
<body>

<h1 style="color:blue;">A Blue Heading</h1>

<p style="color:red;">A red paragraph.</p>

</body>
</html>
```

Internal CSS

- An internal CSS is used to define a style for a single HTML page.
- An internal CSS is defined in the <head> section of an HTML page, within a <style> element.

```
<!DOCTYPE html>
<html>
<head>
<style>
body {background-color: powderblue;}
h1   {color: blue;}
p    {color: red;}
</style>
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

External CSS

- An external style sheet is used to define the style for many HTML pages.
- To use an external style sheet, add a link to it in the `<head>` section of each HTML page
- The external style sheet can be written in any text editor. The file must not contain any HTML code, and must be saved with a `.css` extension.
- `Style.css` is as in the example given
- With an external style sheet, you can change the look of an entire web site, by changing one file!

```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="styles.css">
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

"styles.css":

```
body {
  background-color: powderblue;
}
h1 {
  color: blue;
}
p {
  color: red;
}
```

Link to External CSS

- External style sheets can be referenced with a full URL or with a path relative to the current web page.

```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="styles.css">
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="/html/styles.css">
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

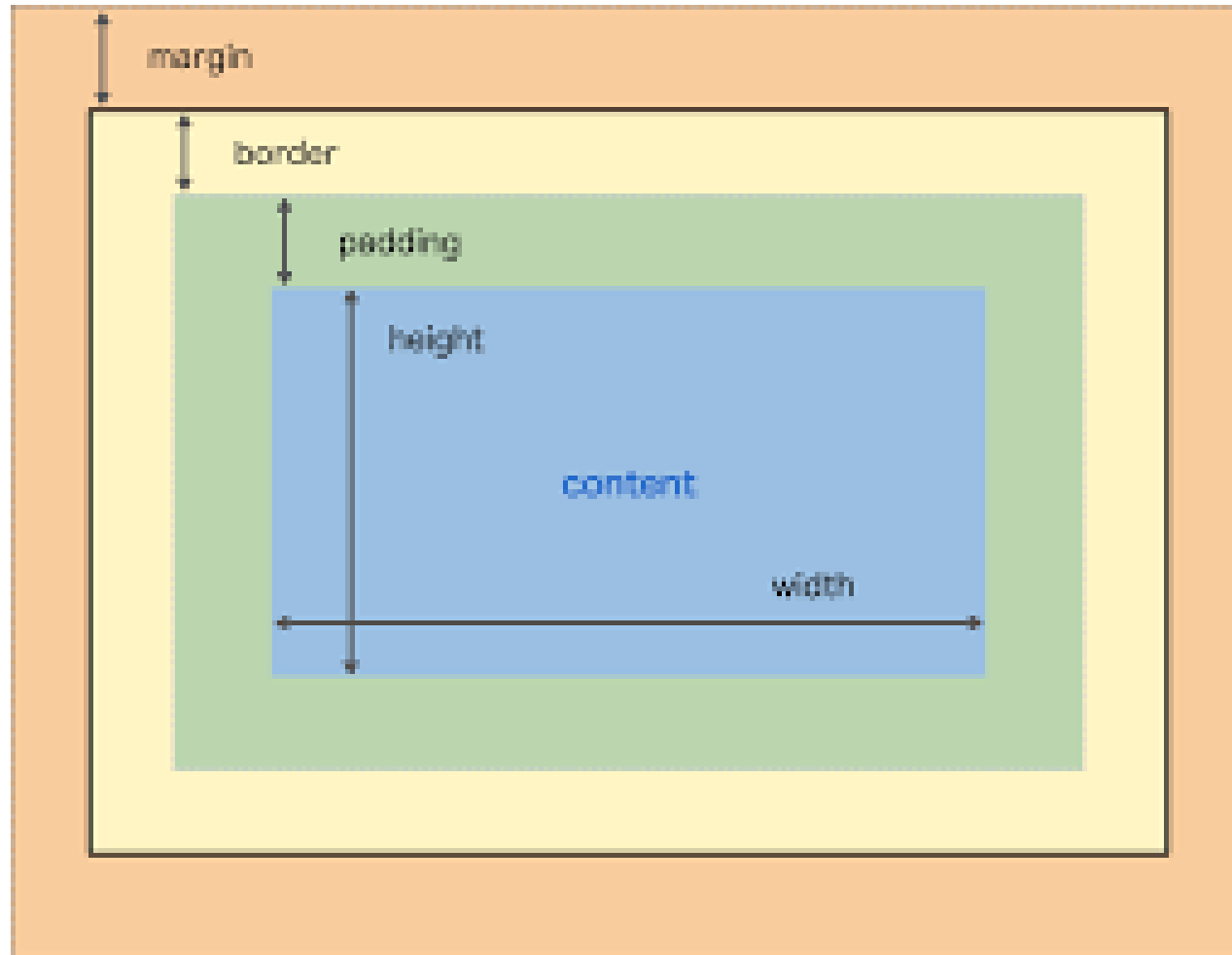
</body>
</html>
```

The CSS Box Model

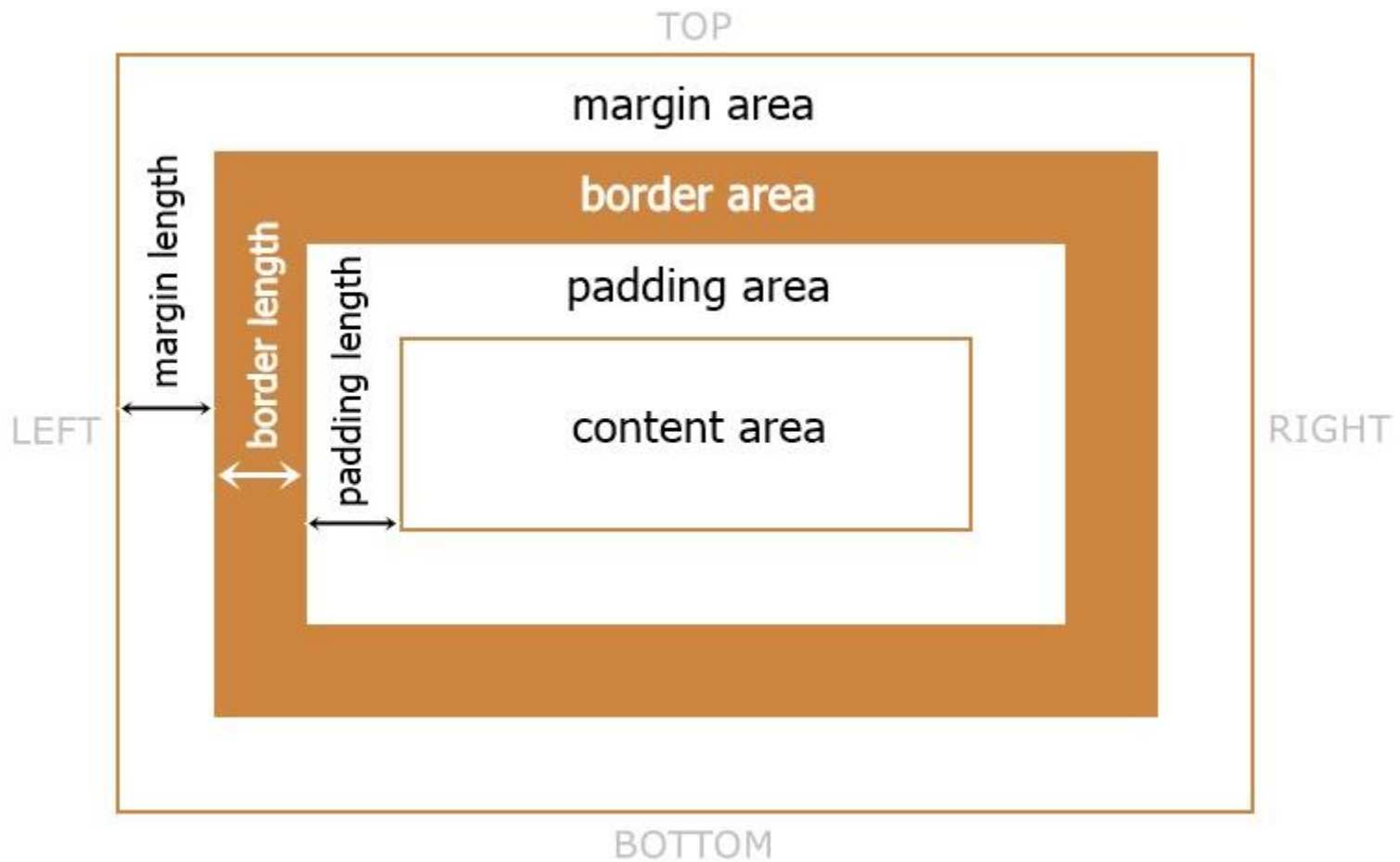
- Every element on a webpage is a rectangular box, composed of four layers from the innermost to the outermost:
- **Content:** The actual content of the element (e.g., text, an `<img>` tag).
- **Padding:** Transparent space around the content, *inside* the border. It inherits the background color of the content area.
- **Border:** A visible line that goes around the padding and content.
- **Margin:** Transparent space *outside* the border, separating the element from other adjacent elements.

CSS Box Model

CSS Border, Padding, Margin



CSS Border Padding & Margin



CSS Border, Padding, Margin

- The CSS border property defines a border around to nearly to all HTML elements.
- The CSS padding property defines a padding (space) between the text and the border.
- The CSS margin property defines a margin (space) outside the border.

CSS Border, Padding

- Border

```
<!DOCTYPE html>
<html>
<head>
<style>
p {
  border: 2px solid powderblue;
}
</style>
</head>
<body>

<h1>This is a heading</h1>

<p>This is a paragraph.</p>
<p>This is a paragraph.</p>
<p>This is a paragraph.</p>

</body>
</html>
```

- padding

```
<!DOCTYPE html>
<html>
<head>
<style>
p {
  border: 2px solid powderblue;
  padding: 30px;
}
</style>
</head>
<body>

<h1>This is a heading</h1>

<p>This is a paragraph.</p>
<p>This is a paragraph.</p>
<p>This is a paragraph.</p>

</body>
</html>
```

CSS Margin

```
<!DOCTYPE html>
<html>
<head>
<style>
p {
  border: 2px solid powderblue;
  margin: 50px;
}
</style>
</head>
<body>

<h1>This is a heading</h1>

<p>This is a paragraph.</p>
<p>This is a paragraph.</p>
<p>This is a paragraph.</p>

</body>
</html>
```

CSS Colors, Fonts and Sizes

- The CSS color property defines the text color to be used.
- The CSS font-family property defines the font to be used.
- The CSS font-size property defines the text size to be used.

```
<!DOCTYPE html>
<html>
<head>
<style>
h1 {
  color: blue;
  font-family: verdana;
  font-size: 300%;
}
p {
  color: red;
  font-family: courier;
  font-size: 160%;
}
</style>
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

CSS Selector

HTML id selector

- The HTML id attribute is used to specify a unique id for an HTML element.
- You cannot have more than one element with the same id in an HTML document. That is, the value of the id attribute must be unique within the HTML document.
- The id attribute is used to point to a specific style declaration in a style sheet. It is also used by JavaScript to access and manipulate the element with the specific id.
- **syntax for id:** write a hash character (#), followed by an id name. Then, define the CSS properties within curly braces {}.
- The id name is case sensitive!
- The id name must contain at least one character, cannot start with a number, and must not contain whitespaces (spaces, tabs, etc.).

example

```
<!DOCTYPE html>
<html>
<head>
<style>
#myHeader {
  background-color: lightblue;
  color: black;
  padding: 40px;
  text-align: center;
}
</style>
</head>
<body>

<h2>The id Attribute</h2>
<p>Use CSS to style an element with the id "myHeader":</p>

<h1 id="myHeader">My Header</h1>

</body>
</html>
```

HTML class Selector

- The HTML class selector is used to specify a class for an HTML element.
- Multiple HTML elements can share the same class.
- The class attribute is often used to point to a class name in a style sheet.
- It can also be used by a JavaScript to access and manipulate elements with the specific class name.
- **Syntax:** To create a class; write a period (.) character, followed by a class name. Then, define the CSS properties within curly braces {}

Class example

```
<!DOCTYPE html>
<html>
<head>
<style>
.city {
  background-color: tomato;
  color: white;
  border: 2px solid black;
  margin: 20px;
  padding: 20px;
}
</style>
</head>
<body>

<div class="city">
<h2>London</h2>
<p>London is the capital of England.</p>
</div>

<div class="city">
<h2>Paris</h2>
<p>Paris is the capital of France.</p>
</div>

<div class="city">
<h2>Tokyo</h2>
<p>Tokyo is the capital of Japan.</p>
</div>

</body>
</html>
```

Example-class

- In the following example we have two `<span>` elements with a class attribute with the value of "note".
- Both `<span>` elements will be styled equally according to the `.note` style definition in the head section
- The class name is case sensitive!

Class example

```
<!DOCTYPE html>
<html>
<head>
<style>
.note {
  font-size: 120%;
  color: red;
}
</style>
</head>
<body>

<h1>My <span class="note">Important</span> Heading</h1>
<p>This is some <span class="note">important</span> text.</p>

</body>
</html>
```

Class example

```
<!DOCTYPE html>
<html>
<head>
<style>
.city {
  background-color: tomato;
  color: white;
  padding: 10px;
}
</style>
</head>
<body>

<h2>The class Attribute</h2>
<p>Use CSS to style elements with the class name "city":</p>

<h2 class="city">London</h2>
<p>London is the capital of England.</p>

<h2 class="city">Paris</h2>
<p>Paris is the capital of France.</p>

<h2 class="city">Tokyo</h2>
<p>Tokyo is the capital of Japan.</p>

</body>
</html>
```

Multiple Classes

- HTML elements can belong to more than one class.
- To define multiple classes, separate the class names with a space, e.g. `<div class="city main">`. The element will be styled according to all the classes specified.
- In the following example, the first `<h2>` element belongs to both the city class and also to the main class, and will get the CSS styles from both of the classes

Example-multiple class

```
<!DOCTYPE html>
<html>
<head>
<style>
.city {
  background-color: tomato;
  color: white;
  padding: 10px;
}

.main {
  text-align: center;
}
</style>
</head>
<body>

<h2>Multiple Classes</h2>
<p>Here, all three h2 elements belongs to the "city" class. In addition, London also
belongs to the "main" class, which center-aligns the text.</p>

<h2 class="city main">London</h2>
<h2 class="city">Paris</h2>
<h2 class="city">Tokyo</h2>

</body>
</html>
```

Different Elements Can Share Same Class

- Different HTML elements can point to the same class name.
- In the following example, both `<h2>` and `<p>` point to the "city" class and will share the same style
- The class name can also be used by JavaScript to perform certain tasks for specific elements.
- JavaScript can access elements with a specific class name with the `getElementsByClassName()` method

example

```
<!DOCTYPE html>
<html>
<head>
<style>
.city {
  background-color: tomato;
  color: white;
  padding: 10px;
}
</style>
</head>
<body>

<h2>Different Elements Can Share Same Class</h2>

<p>Even if the two elements do not have the same tag name, they can both point to the
same class, and get the same CSS styling:</p>

<h2 class="city">Paris</h2>
<p class="city">Paris is the capital of France.</p>

</body>
</html>
```

Example

A class name can be used by multiple HTML elements, while an id name must only be used by one HTML element within the page

```
<!DOCTYPE html>
<html>
<head>
<style>
/* Style the element with the id "myHeader" */
#myHeader {
  background-color: lightblue;
  color: black;
  padding: 40px;
  text-align: center;
}

/* Style all elements with the class name "city" */
.city {
  background-color: tomato;
  color: white;
  padding: 10px;
}
</style>
</head>
<body>

<h2>Difference Between Class and ID</h2>
<p>A class name can be used by multiple HTML elements, while an id name must only be used by one HTML element within the page:</p>

<!-- An element with a unique id -->
<h1 id="myHeader">My Cities</h1>

<!-- Multiple elements with same class -->
<h2 class="city">London</h2>
<p>London is the capital of England.</p>

<h2 class="city">Paris</h2>
<p>Paris is the capital of France.</p>

<h2 class="city">Tokyo</h2>
<p>Tokyo is the capital of Japan.</p>

</body>
</html>
```

Types of Selectors

```
<!DOCTYPE html>
<html>
<head>
<style>
```

```
/* ----- 1. Element Selector & element-child selector----- */
```

```
p {
  color: navy;
}
```

```
h1 b{
  background-color: yellow;
}
```

```
/* ----- 2. Class Selector ----- */
```

```
.highlight {
  background-color: yellow;
}
```

```
p.box1 {
  border: 2px solid blue;
}
```

```
/* ----- 3. ID Selector ----- */
```

```
#main-title {
  color: darkred;
  text-align: center;
}
```

```
/* ----- 4. Group Selector ----- */
```

```
h2, h3 {
  color: green;
}
```

```
/* ----- 5. Descendant Selector ----- */
```

```
.box p {
  color: purple;
}
```

```
/* ----- 6. Child Selector ----- */
```

```
.box > span {  
|   color: orange;  
}
```

```
/* ----- 7. Adjacent Sibling Selector ----- */
```

```
h3 + p {  
|   font-style: italic;  
}
```

```
/* ----- 8. General Sibling Selector ----- */
```

```
h3 ~ ul {  
|   border: 2px dashed gray;  
}
```

```
/* ----- 14. Universal Selector ----- */
```

```
* {  
|   font-family: Arial, sans-serif;  
}
```

```
</style>|
```

```
</head>
```

✓ <body>

```
<h1 id="main-title">ALL CSS SELECTORS DEMO</h1>
```

```
<h2>Element & Class Selector</h2>
```

```
<p>This is a normal paragraph</p>
```

```
<p class="highlight">This paragraph uses class selector</p>
```

```
<h3>Descendant & Child Selector</h3>
```

✓ <div class="box">

```
  <p>Paragraph inside div (descendant)</p>
```

```
  <span>Span inside div (child)</span>
```

```
</div>
```

```
<h3>Sibling Selectors</h3>
```

```
<p>This paragraph is adjacent sibling</p>
```

✓

```
  <li>Item One</li>
```

```
  <li>Item Two</li>
```

```
  <li>Item Three</li>
```

```
</ul>
```

```
<p class="box1"> Box is only applied to p</p>
```

```
<div class="box1"><p> this is box p</p></div>
```

```
<h1> this is <b> bold text inside heading</b></h1>
```

```
</body>
```

```
</html>
```

HTML Bookmarks with ID and Links

- HTML bookmarks are used to allow readers to jump to specific parts of a webpage.
- Bookmarks can be useful if your page is very long.
- To use a bookmark, you must first create it, and then add a link to it.
- Then, when the link is clicked, the page will scroll to the location with the bookmark.

Example

- First, create a bookmark with the id attribute:
`<h2 id="C4">Chapter 4</h2>`
- Then, add a link to the bookmark ("Jump to Chapter 4"), from within the same page:

```
<!DOCTYPE html>
<html>
<head>
  <title>HTML Bookmarks Example</title>

  <style>
    body {
      font-family: Arial;
    }

    .menu {
      background-color: lightblue;
      padding: 10px;
      text-align: center;
    }

    .menu a {
      margin: 10px;
      text-decoration: none;
      color: blue;
      font-weight: bold;
    }

    .section {
      height: 500px; /* to enable scrolling */
      padding: 20px;
    }
  </style>
</head>
```

```
<body>
```

```
  <!-- Bookmark Links -->
```

```
  <div class="menu">
```

```
    <a href="#html">HTML</a>
```

```
    <a href="#css">CSS</a>
```

```
    <a href="#js">JavaScript</a>
```

```
  </div>
```

```
  <!-- Bookmark Targets -->
```

```
  <div class="section" id="html">
```

```
    <h2>HTML Section</h2>
```

```
    <p>This is the HTML section.</p>
```

```
  </div>
```

```
  <div class="section" id="css">
```

```
    <h2>CSS Section</h2>
```

```
    <p>This is the CSS section.</p>
```

```
  </div>
```

```
  <div class="section" id="js">
```

```
    <h2>JavaScript Section</h2>
```

```
    <p>This is the JavaScript section.</p>
```

```
  </div>
```

```
</body>
```

```
</html>
```